Spatial Gap-Filling of SMAP Soil Moisture Pixels Over Tibetan Plateau via Machine Learning Versus Geostatistics

Cheng Tong, Hongquan Wang, Ramata Magagi, *Member, IEEE*, Kalifa Goïta, *Member, IEEE*, and Ke Wang, *Member, IEEE*

***Abstract*—Soil moisture (SM) is a key variable in ecology, environment, agriculture, and hydrology. The Soil Moisture Active Passive (SMAP) satellite provides global SM products with reliable accuracy since 2015. However, significant gaps of SMAP SM appeared over Tibetan Plateau. Considering the important role of the Tibetan Plateau in global climate and environment, it is essential to develop methods to infill the gaps to generate seamless SMAP SM data. To address this issue, we proposed two methods, machine learning and geostatistics technique. For the machine learning technique, we train a Random Forest algorithm which aims to match the output of available SMAP L3 SM using a series of input variables such as SMAP brightness temperature (TB_H and TB_V) in ascending orbits (6:00 PM local time), surface temperature, MODIS NDVI, land cover, DEM, and other auxiliary data. Then, the established RF estimators were applied to the SMAP brightness temperature from descending orbits (6:00 AM local time) to reconstruct complete SM data over the Tibetan Plateau. For the geostatistics technique, the Ordinary kriging was applied to the available SMAP L3 SM pixels to interpolate complete SM data. To cross-validate the performances of the algorithms, we assume certain areas with available SMAP SM values as missing, and then compared the gap-filling results with the actual ones. The cross-validations show that the gap-filling results from two algorithms were highly correlated to the official SMAP SM products with high coefficients of determination ($R^2{}_{\mathrm{RF}} = 0.97$ and $R^2{}_{\mathrm{OK}} = 0.85$) and low RMSE ($\mathrm{RMSE}_{\mathrm{RF}} = 0.015$ cm^3/cm^3 and $\mathrm{RMSE}_{\mathrm{OK}} = 0.036$ cm^3/cm^3). Furthermore, the gap-filling SM data present a better correlation with the Soil Moisture and Ocean Salinity SM data ($R = 0.55$–0.7) than the Global Land Data Assimilation System simulations ($R = 0.18$–0.62). The reconstructed SM from RF ($R = 0.71$) and OK ($R = 0.55$) algorithms are well related to the Maqu network measurements. Thus, the machine learning and geostatistics algorithms have the potential to reproduce the missing SMAP SM products over the Tibetan Plateau.**

***Index Terms*—Gap-filling, geostatistics, machine learning, soil moisture (SM), soil moisture active passive (SMAP), tibetan plateau.**

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I. Introduction

Soil moisture (SM) is a significant environmental variable that controls surface water circulation and energy exchanges between the atmosphere and land surface [1]. It also plays a key role in the fields of ecology, environment, agriculture, hydrology, and meteorology [2]–[4]. On a regional scale, estimating land SM contributes to a knowledge of ecological response to global climate change [5]. For human daily production and life, an accurate description of SM provides an important signal for weather forecasting, field water resource management, disaster (drought and flood) warning, crop growth monitoring, and yield estimation [6]–[8]. Therefore, estimating SM accurately is indispensable to meet scientific and practical requirements.

With the capacity of penetrability, microwave remote sensing sensors work on all-weather and day/night conditions, and thus provide a promising technique to estimate long time series SM at the regional and global scales [9]. Currently, with greater temporal resolution and lower interference of surface roughness than the active microwave, the passive microwave has been applied to estimate global SM with daily observation. A series of passive microwave SM products such as Advanced Microwave Scanning Radiometer 2 (AMSR2), Soil Moisture and Ocean Salinity (SMOS), Soil Moisture Active Passive (SMAP), and Fengyun-3B provide freely adequate data sources for various applications [10], [11]. Particularly, the SMAP mission was originally designed to incorporate L-band radiometer and radar data to retrieve SM with unbiased Root Mean Square Error (ubRMSE) $\leq$ 0.04 cm^3/cm^3 [12]. Since the malfunction of SMAP radar on July 7, 2015, only the SMAP radiometer operated and provided 36 km brightness temperature observations and SM products at 6:00 am (descending)/6:00 pm (ascending) local time with 2–3 days global coverage.

The SMAP SM products were retrieved at different spatial resolutions (36, 9, 3, and 1 km) from brightness temperature and other auxiliary data. Extensive validation works showed that the SMAP SM products performed well in comparison to *in-situ* measurements [13]–[16]. For example, Zeng *et al.* [13] evaluated the SMAP SM products against *in-situ* measurements collected from three networks across the U.S., Finland, and Romania. The results showed the SMAP products agreed with *in-situ* measurements, with a correlation coefficient of 0.7 and

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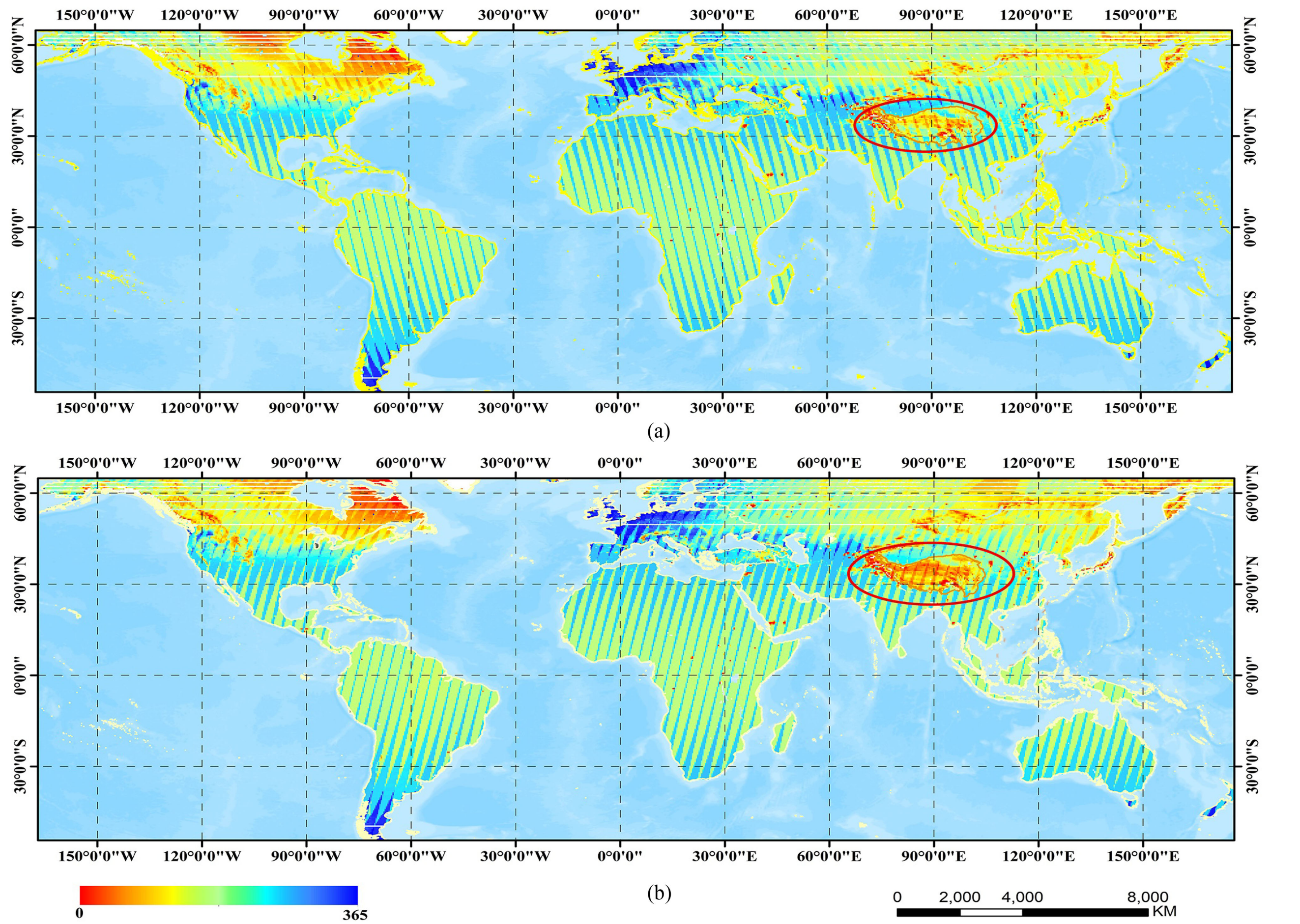

Fig. 1. Temporal coverage (available days in 2018) of global SMAP retrieved soil moisture in (a) ascending and (b) descending orbits.

an overall ubRMSE of 0.036 cm^3/cm^3. Compared with other satellite-derived SM, some studies demonstrated that the SMAP SM products have greater performances at a global scale [17], [18]. Ma *et al.* [18] evaluated several remotely sensed SM data including SMOS, AMSR2, SMAP, and ESA climate change initiative (CCI), by comparison with global ground-based measurements. Their results showed that the SMAP data captured the temporal trends of ground SM, exhibiting the highest correlation coefficient of 0.73 at a global scale.

The Tibetan Platea is known as "the roof of the World" and "water tower of Asia," with an average altitude of over 4000 m. Due to its unique geographical location and geomorphologic features, the Tibetan Plateau plays a key role in the formation and evolution of the Asian monsoon, and even has an important influence on the global atmospheric water cycle. Thus, the timely information on the SM over the Tibetan Plateau is helpful to optimize the regulations of the local ecological environment and climate change. However, over the Tibetan Plateau region, the availability of SMAP SM is significantly fewer than that in other inland regions. For example, for half of the time in 2018, the SMAP SM data were absent for most areas of the Tibetan Plateau (delineated in a red ellipse in Fig. 1). Considering the importance of the Tibetan Plateau in the global environment, it is necessary to develop gap-filling strategies to obtain temporally and spatially complete SMAP SM data.

To address the issue of pixel missing, various gap-filling methods were proposed to reconstruct the missed SM data. For example, to fill the missing values of ECV (essential climate variable) SM pixels in Europe, Liu *et al.* [19] used valid SMAP SM by calculating their arithmetic means to fill the missing ECV SM and enhance the spatial coverage of ECV. However, this gap-filling strategy mainly depends on available SMAP pixels, and when SMAP data was not available, it was not applicable. Zhang *et al.* [20] proposed an artificial neural network algorithm to reconstruct the missed data of ESA CCI from 1982 to 2015 across China by using numerous environmental variables as training data. In our previous work, we compared the ordinary least squares, support vector machine, and Random Forest (RF) methods to estimate SM from SMAP brightness temperature [21]. The estimated SM maintained high consistency with SMAP SM, and the RF model was found to obtain the best performance. Thus, we assumed that the RF algorithm may provide a solution to fill the data gap when the SMAP SM products were missing. On the other hand, geostatistical techniques which take advantage of spatial autocorrelation of variables were applied in remote sensing data gap-filling successfully [22]–[24]. Llamas

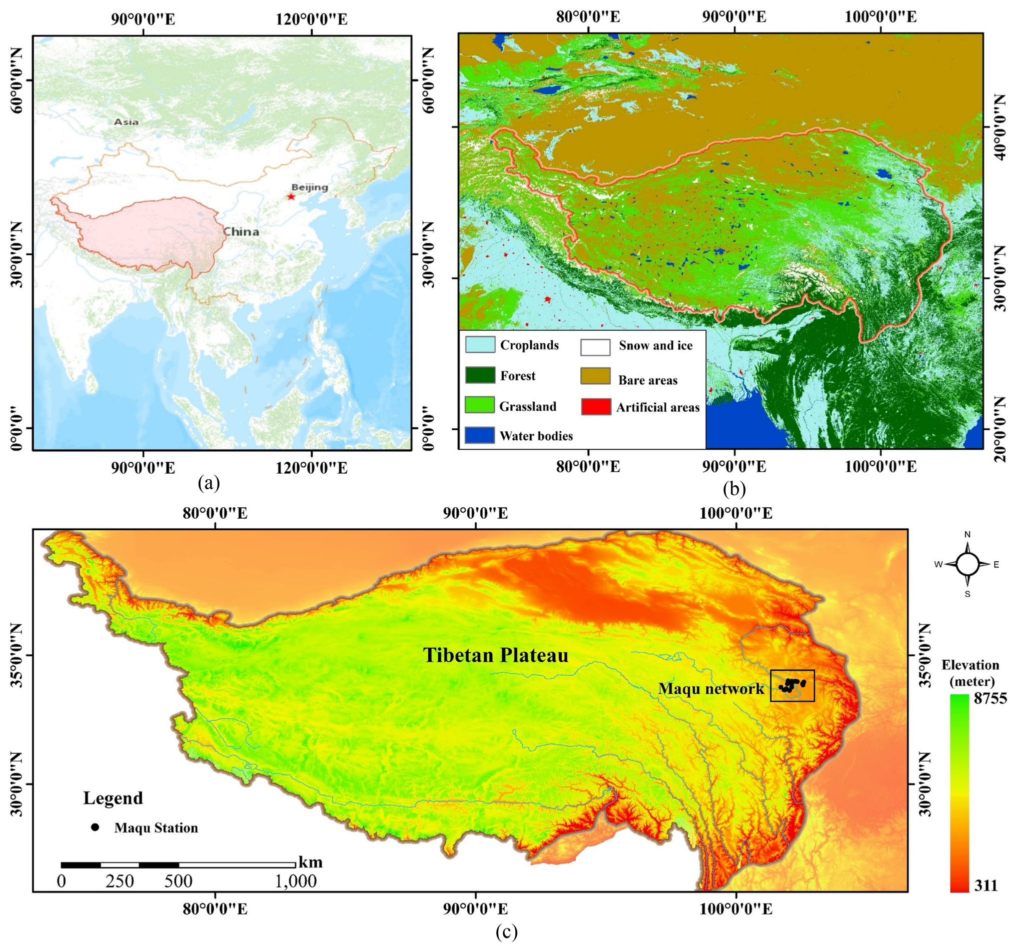

Fig. 2. Study area (a) location, (b) land cover, and (c) topography.

et al. [24] evaluated Ordinary kriging (OK), regression kriging, and generalized linear models to fill spatial gaps of ESA CCI SM. The results revealed that the geostatistics technique could provide an alternative method to reproduce the missing values of satellite-derived SM.

Considering the large pixel missing in the SMAP retrieved SM over the Tibetan Plateau, our current article proposes machine learning algorithms versus geostatistics models to reconstruct the gaps. In the following, Section II introduces the datasets over the study area and then develops two different SMAP SM gap-filling schemes. Section III analyzes the results, followed by the discussions in Section IV. The conclusions are summarized in Section V.

II. Materials and Methodology

A. Study Area

The study site is located in the Tibetan Plateau (73°–104°E, 25°–40°N). As shown in Fig. 2, there is a uniform land cover type, mainly distributed alpine meadow. Due to the influence of high altitude and the Asian monsoon, the climate belongs to cold semiarid, precipitation is mainly concentrated in summer from June to August. During the winter, very little precipitation occurred and with annual freeze/thaw cycle. Usually, soil surface starts to freeze around October, until the occurrence of thawing in the following January of the second year. Owing to harsh climatic conditions, fewer human activities lead to little variation in soil roughness.

B. Datasets

1) SMAP Brightness Temperature Observations: The SMAP satellite provides daily TB observations from March 31st, 2015 to nowadays. In this article, we select SMAP Level-3 brightness temperature (TB) product (SPL3TB, version 6), which is gridded on a 36 km grids at cylindrical equal-area scalable earth (EASE) grid version 2 [25], for both descending (around 6:00 am local time) and ascending (6:00 pm) overpasses. Compared with SMAP Level-1C TB observations, the SMAP L3 TB products compute the arithmetic means of the same variables provided in fore- and aft-looking groups in the input L1C_TB granule, and are adjusted for the presence of water fraction by a threshold [26]. However, owing to satellite scanning gaps, the available observation numbers of individual pixels from SMAP in the orbit gap are fewer than in other areas. In this article, we use

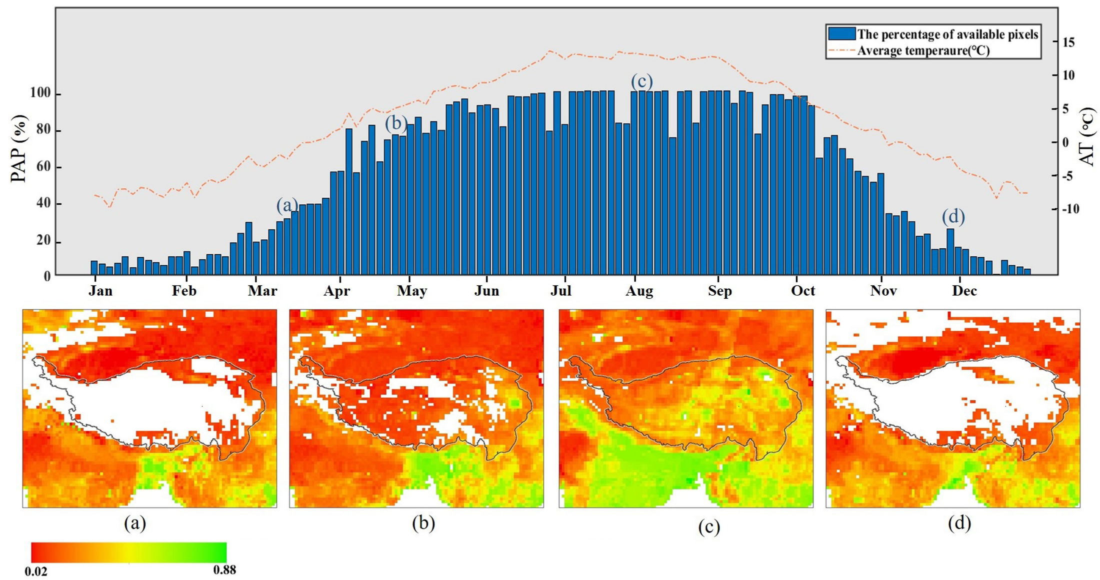

Fig. 3. Temporal variations of available SM pixels in (a) March, (b) April, (c) August, and (d) December in 2018.

brightness temperature at horizontal polarization (TB_H) and vertical polarization (TB_V) for the gap-filling algorithms.

2) SMAP L3 SM Product: The daily SMAP L3 SM product used in this article is retrieved by the V-pol Single Channel Algorithm with SMAP TB and other auxiliary data including surface temperature (ST), vegetation water content, and surface roughness information [27]. The SMAP L3 SM products covered the study area at a revisit interval of 3 days. Owing to the strict requirements on the external conditions, the SMAP SM retrieval is not feasible under some conditions: 1) vegetation water content (VWC)$\geq$5 kg/m^2; 2) ST $<$273 K; or 3) urban area and water body [28]. In addition to satellite scanning gaps (Section II-B.1), such conditions increase the pixel missing issue in the SMAP SM data.

Over the Tibetan Plateau, available SMAP SM pixels were not stable in space and time from 2015 to nowadays. In this article, we chose the SMAP data in 2018 to develop and evaluate the gap-filling algorithms for the SMAP missing pixels. To quantify the degrees of missing pixels, we defined an index, namely percentage of available pixels (PAP):

$$\mathrm{PAP} = \frac{N_i}{N_{\mathrm{all}}} \times 100\% \tag{1}$$

where N_{all} and N_i represents the amount of all pixels in the study area and the number of available pixels in the ith day, respectively.

Fig. 3 indicates the temporal variation of PAP over the Tibetan Plateau in 2018. From January to April, more than 50% of pixels were missed. It is important to deepen the understandings of the factors which may cause the data missing. With the increase of average temperature, the number of available data increases. Thus, the soil temperature is assumed to be a major factor influencing the number of available pixels. Furthermore, in terms of the SMAP baseline algorithm, the presence of water bodies, urban areas, dense vegetation (VWC $\geq$ 5 kg/m^2), and low ST (ST $<$ 273K) were the main factors affecting SMAP retrieval. In theory, vegetation was flourishing during summer from July to August over Tibetan Plateau. However, during this period, we observed few missing SMAP SM pixels. Thus, the vegetation imposed marginal interference on the retrieval rate, but it may impact the retrieval accuracy which is beyond the scope of our current article. Consequently, the ST is a major factor influencing the availability of valid SMAP SM pixels over the Tibetan Plateau.

From the analysis of the spatial relationship between the ST and the number of valid pixels during different seasons of 2018, four cases were summarized in Fig. 4. In addition, Fig. 5 shows the temporal variation of the percentages of these cases through that year. From November to April of the next year, the absence of SMAP SM pixels was related to low temperature (below 273 K) and frozen soils which is not considered in this article. However, when the ST was above 273 K from May to October, SMAP SM was still missing over some areas. For this condition, the missing pixels were probably related to other factors such as radio frequency interference (RFI) contamination, and it was thus relevant to reconstruct the missed SM. Meanwhile, it was worthy to note a situation where the local ST was below 273K but a small amount of SM still existed. This anomaly situation did not meet the SMAP baseline algorithm, and may be caused by system errors or retrieval uncertainty.

3) Modis NDVI: The MODIS sensor provides a series of land parameters products such as vegetation index, land ST, and land cover for scientific applications. This article used the MOD13Q1 NDVI data in sinusoidal projection mode, 16-day temporal resolution, and 250 m spatial resolution. Due to the contamination of clouds, aerosols, and cloud shadow, the NDVI products are reprocessed by sifting the optimum available pixels

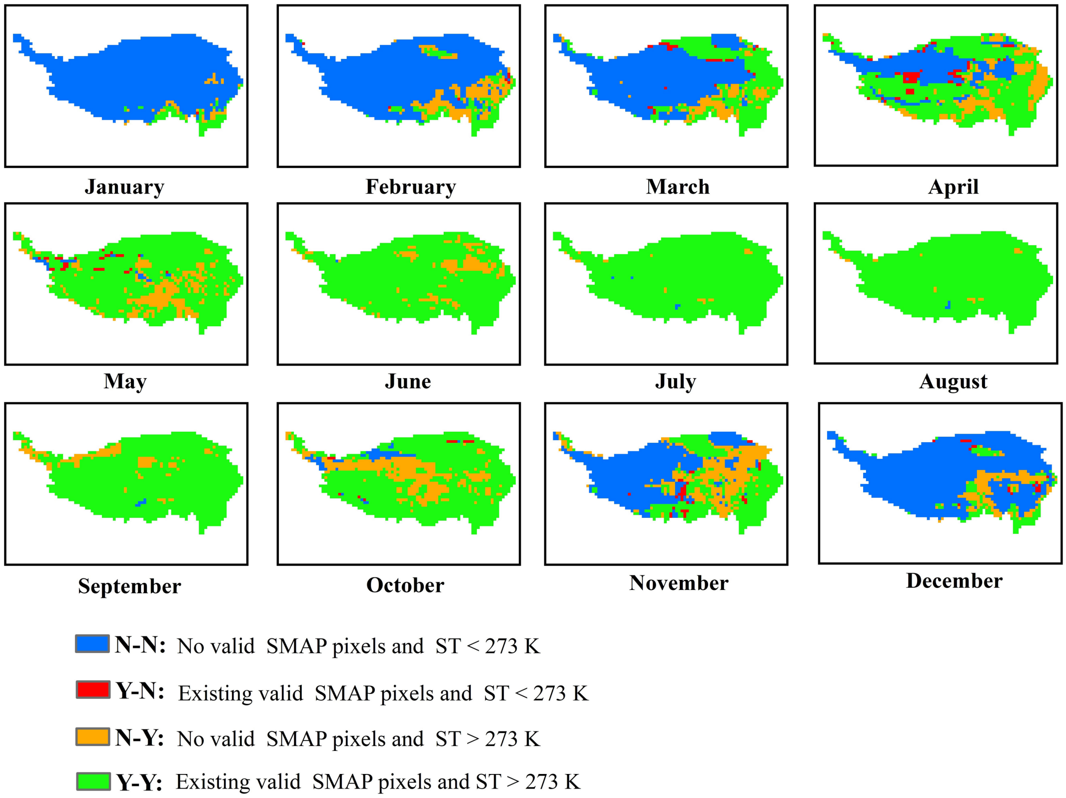

Fig. 4. Spatial relationship between surface temperature and number of valid SMAP SM pixels over Tibetan Plateau in 2018.

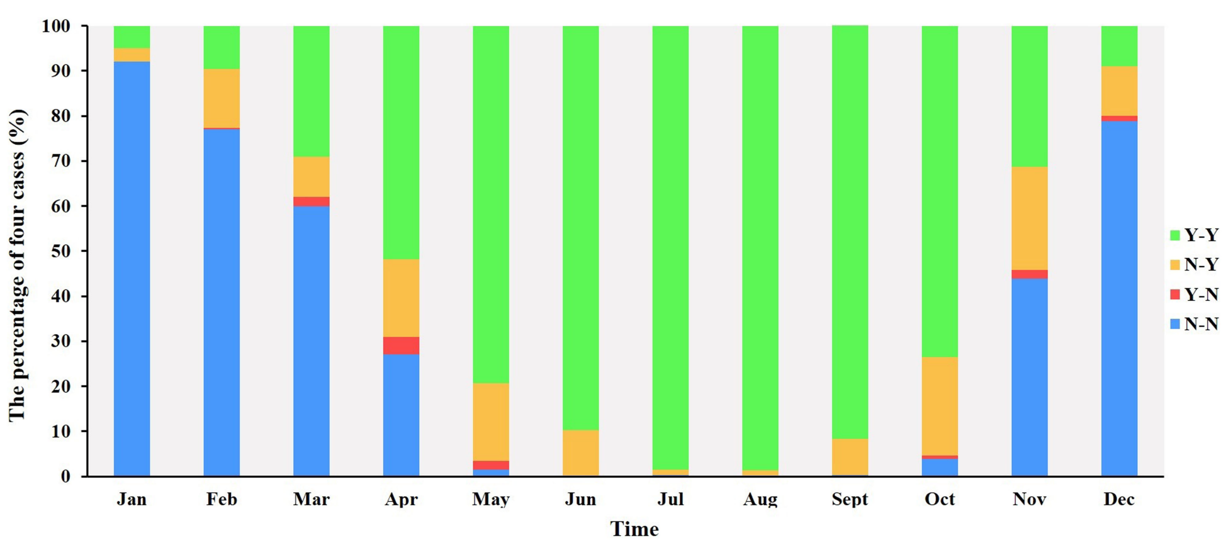

Fig. 5. Temporal profile of the percentages of four pixel-missing cases (Fig. 4) over Tibetan Plateau in 2018.

from all the original acquisitions during 16 days period for atmosphere correction, cloud-filtered, and quality check [29]. All the NDVI pixels are aggregated to 36 km as the SMAP pixels, and reprojected to the WGS84 coordinate system using the MODIS reprojection tool.

4) Auxiliary Data: Aside from the optical and microwave data, we also utilized other datasets such as ST, DEM, and land cover. The daily ST data is from NASA's GEOS-5 forward processing atmospheric data assimilation system. Compared with other ST data such as MODIS data, GEOS-5 ST has more completely spatial and temporal coverage. The GEOS-5 ST data were aggregated to the same 36-km EASE grid as the SMAP L3 product. As for land cover data, we selected the MODIS MCD12Q1 at 500 m spatial resolution, with 17 main types of land covers. The DEM data were acquired from Shuttle Radar Topography Mission.[1]

5) Other Soil Moisture Datasets: To further evaluate the quality of the gap-filling results, we also used the Global Land Data Assimilation System (GLDAS) and SMOS L3 SM datasets for comparison. The GLDAS is to generate a series of land surface meteorological data including land ST, air pressure, wind speed, soil water content, snow equivalent, and runoff [30]. The GLDAS SM data were obtained by using the land surface model with forcing data from satellite- and ground-based observational data, and provided complete temporal and spatial distribution without obvious data missing circumstance [31]. In this article, the GLDAS Noah SM product with a temporal resolution of 3 h and 0.25° spatial resolution at the depth of 10 cm is used. Meanwhile, the daily SMOS Level-3 SM product with a spatial resolution of 25 km is also used for comparison. The SMOS mission was designed to monitor global surface SM via a passive microwave inversion algorithm with an accuracy of 0.04 cm^3/cm^3 [32]. Since both the SMAP and SMOS used radiometer at L-band, the intercomparison of the SMAP gap-filling results with the SMOS retrieved SM provide a way to characterize the robustness of the algorithms. The GLDAS and SMOS data are resampled to a 36 km grid size to match the SMAP pixels.

Furthermore, the *in-situ* SM data were acquired with 15 min intervals in 2018, at different depths (5, 10, 20, 40, and 80 cm) over 15 stations of the Maqu networks located in the east of the Tibetan Plateau [33], [34]. Considering the penetrating depth of microwave at L-band, this article adopted the portion of the SM data at 5 cm to evaluate the gap-filling SMAP SM.

C. Methodology

To reconstruct the missed SMAP SM pixels over the Tibetan Plateau, this article proposes machine learning and geostatistical algorithms as illustrated in Fig. 6. They are further interpreted as follows:

1) RF Algorithm: The RF approach, as an ensemble machine learning algorithm, has been widely used in SM retrieval and reconstruction [35], [36]. Previous studies compared the performances of different machine learning approaches in the retrieval of SM, indicating that the RF model obtained robust results [21], [37], [38]. The idea of RF is to train multiple decision trees for prediction. Each decision tree is relatively independent and has a predicted value for a variable, and the final result is the average estimates from all decision trees. The performance of the RF model is highly sensitive to the number of adopted decision trees which is calibrated in this article. In contrast, other parameters such as maximum split size, maximum node size, and minimum leaf size were set to default values in the calibration process [39]–[41].

Since there were more SM missing pixels in the descending orbits than in the ascending orbits (Fig. 1), the SMAP ascending TB and the simultaneous auxiliary data are selected to train an estimator. As shown in Fig. 6, the input variables for the RF algorithm were composed of SMAP brightness temperature (TB_H and TB_V), GEO-5 ST, MODIS NDVI, land cover, and DEM, while the output variable was SMAP L3 SM. All training data were adopted during the unfrozen periods (ST > 273 K). For the RF algorithm, we utilized a five-fold cross-validation technique to test the performances of the gap-filling. The entire training data were randomly classified into five groups: four groups were selected to establish an estimator, and the remaining one group was used as test data. Repetitive work continued five times, and their average performance score represents the overall results. In the process of cross-validation, the test data could be assumed as missing data. Then, these test data were used to validate the performance of the gap-filling values. After the cross-validation, the descending data are used to reconstruct the missed SMAP SM retrieval.

2) OK Algorithm: The OK is a typical geostatistical method that provided linear unbiased predictions. For the SMAP SM with data gaps, we adopted the OK method to interpolate complete SM across the Tibetan plateau. First, we assumed a regionalized variable $z(x)$ to be a random process, with the hypothesis that its average is an unknown constant. Given n SMAP SM data $z(x_i)$ at locations i ($i = 1, 2, \ldots, n$), the linear prediction $\mathbf{z}^*(\mathbf{x}_0)$ for a nonsampled location x_0 is expressed as

$$Z^*(x_0) = \sum_{i=1}^{n} \lambda_i z(x_i) \tag{2}$$

where λ_i represents the weigh value with unbiasedness requirement, assigned to the ith SMAP SM. It is calculated as a function of the distance between the ith locations of SMAP SM data and the nonsampled locations (x_0) where SM will be predicted.

The OK method determined the optimal weight value based on unbiased prediction and minimum variance. Thus, the following two conditions must be met to realize the linear unbiased optimal estimation: 1) unbiased condition which represented that the predicted value was consistent with the observed value: $\mathbf{E}\left[\mathbf{z}^*(\mathbf{x}_0) - \mathbf{z}(\mathbf{x}_0)\right] = 0$; and 2) optimum condition which represented that the difference between the predicted value and the observed value has the smallest variance: $\mathbf{Min}(\mathbf{Var}[\mathbf{z}^*(\mathbf{x}_0) - \mathbf{z}(\mathbf{x}_0)])$. For the two conditions, the sum of weight is constrained to be 1.

Compared with other interpolation methods such as regression kriging or cokriging, the OK method does not require any auxiliary input variables but can achieve similar results [24], [42]. Besides, when the autocorrelation (spatial dependence) between the SM is high, the OK method has shown efficiency over the cokriging techniques [43], [44]. Meanwhile, for the OK method, semivariograms $\hat{\gamma}$ are used to characterize the spatial variability of the SMAP sampled SM data [45]:

$$\hat{\gamma} = \frac{1}{2}\frac{1}{N(h)}\sum_{i=1}^{N(h)}\left[Z(S_i + h) - Z(S_i)\right]^2 \tag{3}$$

where $Z(S_i)$ and $Z(S_i + h)$ represent the SMAP SM values at locations S_i and $S_i + h$, respectively. $N(h)$ indicates the number of sample pairs at the step interval h.

[1] [Online]. Available: http://srtm.csi.cgiar.org/srtmdata/

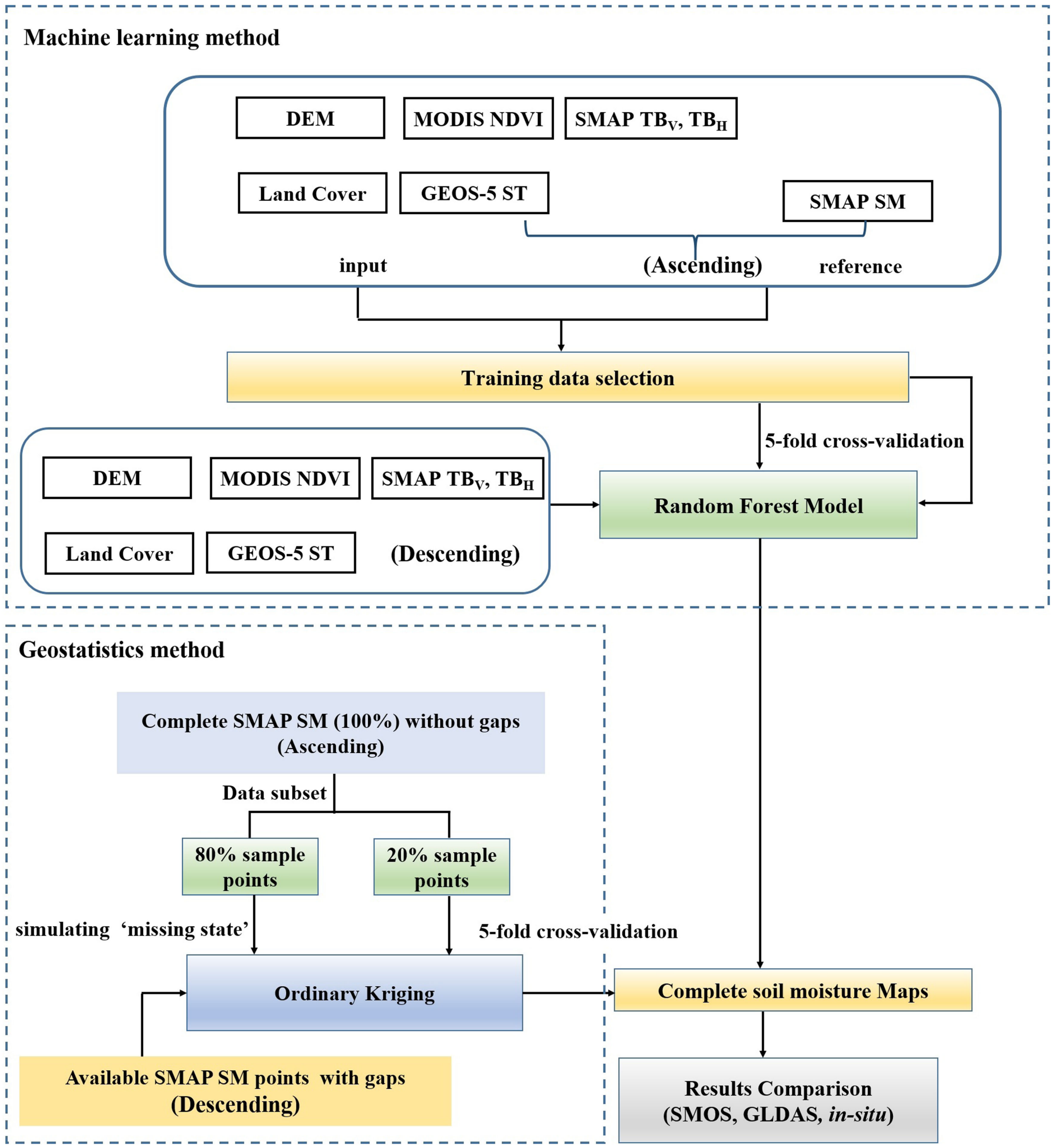

Fig. 6. Diagram of the gap-filling algorithms for missing pixels in SMAP soil moisture.

Since the accuracy of the OK model depends on the semivariograms and associated intrinsic parameters (range, nugget, and sill), we evaluated three of the most common semivariograms: spherical, exponential, and Gaussian. Based on the nugget-to-sill ratio, the spatial dependency index (SDI) [46] is computed to quantify the spatial dependence degree of the semivariogram:

$$\mathrm{SDI} = \frac{C_0}{C_0 + C_1} \times 100\% \tag{4}$$

where C_0 indicates nugget effect, C_1 indicates partial sill, and the sum of C_0 and C_1 represents sill. The conditions of SDI ≤ 0.25, $0.25<$ SDI ≤ 0.75, and SDI > 0.75 indicate strong, moderate, and weak spatial dependence, respectively.

For certain periods during the summer season, there were almost no gaps in the SMAP SM products over the Tibetan Plateau (Fig. 3). This provides a condition to validate the accuracy of the interpolation method. We assumed that 20% of the SMAP pixels at ascending orbits over the study area were missing, and then filled based on the remaining 80% of the SMAP pixels. As such the filled SMAP pixels by the OK algorithm were compared to the assumed missing pixels to calculate statistical metrics for evaluation. Furthermore, similar to the RF algorithm, we conducted the five-fold cross-validation for the OK algorithm. Specifically, 1) 20% of existing SMAP SM pixels were removed randomly, 2) the remaining 80% pixels were used to fill missing data via the geostatistics technique, and 3) the removed 20% pixels were considered as true data to compare with the interpolation results for cross-validation. This process was repeated five times, leading to the validation results. Finally, the established OK algorithm was applied to the SMAP TB at descending orbit to produce complete SM maps over Tibetan Plateau.

3) Quality Evaluation Against Alternative SM Products: To evaluate the performances of the two gap-filling algorithms, this article used five typical evaluation metrics including correlation

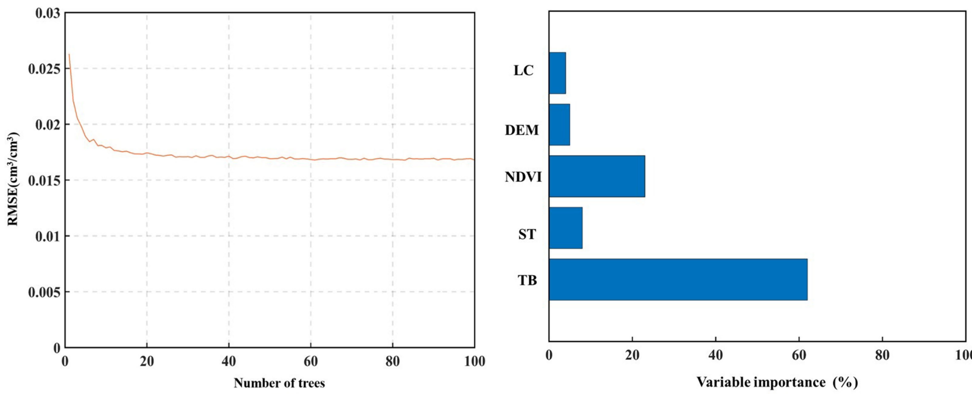

Fig. 7. (a) Variation of RMSE with tree numbers. (b) Relative importance of input variables in RF algorithm.

coefficient (R), root mean square error (RMSE), ubRMSE, bias, and variance:

$$R = \frac{\sum_{i=1}^{N}(\mathrm{SM_{PRE}} - \overline{\mathrm{SM_{PRE}}})(\mathrm{SM_{REF}} - \overline{\mathrm{SM_{REF}}})}{\sqrt{\sum_{i=1}^{N}(\mathrm{SM_{PRE}} - \overline{\mathrm{SM_{PRE}}})^2 \sum_{i=1}^{N}(\mathrm{SM_{REF}} - \overline{\mathrm{SM_{REF}}})^2}} \tag{5}$$

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{i=1}^{N}(\mathrm{SM_{PRE}} - \mathrm{SM_{REF}})^2}{N}} \tag{6}$$

$$\mathrm{ubRMSE} = \sqrt{\frac{\sum_{i=1}^{N}((\mathrm{SM_{PRE}} - \overline{\mathrm{SM_{PRE}}}) - (\mathrm{SM_{REF}} - \overline{\mathrm{SM_{REF}}}))^2}{N}} \tag{7}$$

$$\mathrm{bias} = \frac{\sum_{i=1}^{N}(\mathrm{SM_{PRE}} - \mathrm{SM_{REF}})}{N} \tag{8}$$

$$\mathrm{variance} = \frac{\sum_{i=1}^{N}(\mathrm{SM_{PRE}} - \overline{\mathrm{SM_{PRE}}})^2}{N} \tag{9}$$

where $\mathrm{SM_{PRE}}$ and $\mathrm{SM_{REF}}$ represents, respectively, the predicted value and referenced SM value for validation; $\overline{\mathrm{SM_{PRE}}}$ and $\overline{\mathrm{SM_{REF}}}$ indicate their average values, and N is the number of sample data for comparison.

III. Results

A. Calibration of Random Forest and Ordinary Kriging

Since the performance of RF is strongly related to the number of adopted decision trees, we first optimize the number of decision trees. Fig. 7(a) shows the influence of increasing the number of decision trees on the model's performance in terms of retrieval RMSE. As the number of decision trees increased to 20, the RMSE between predicted and observed values constantly decreased and then became stable with the lowest RMSE. Therefore, 20 decision trees are optimal for the current RF algorithm.

In addition, the RF algorithm provides a relative contribution of each input variable, by calculating the increased mean square error (MSE). The larger the MSE value, the more significant the corresponding input variable. Fig. 7(b) illustrates that the TB and NDVI play important roles in the gap-filling results. This confirms the sensitivity of SMAP TB to SM [21], and also the close correlation between SM and NDVI [47].

The performance of the OK depends on the semivariogram. Fig. 8 shows the SDI and RMSE calculated for three semivariograms. Compared with the other two models, the exponential semivariogram model obtained strong spatial dependence with the lowest SDI and the minimum RMSE. Thus, the exponential semivariogram was selected for the OK model to predict the missing SM values in the following study.

B. Validation of Gap-Filling Results

To validate the gap-filling results, we utilized five-fold cross-validation for the RF model. As for the geostatistical method, we randomly removed 20% of the valid SMAP SM pixels to test the performance. In this cross-validation process, the test data was assumed as missing values, then the remaining data was used to reconstruct the missed pixels. Then, the gap-filling values were compared with the test data.

In Fig. 9(a), the RF predicted filling SM values were compared to the selected test data, indicating a determinant coefficient of 0.97 and an RMSE of 0.015 $\mathrm{cm^3/cm^3}$. Most of the points were distrusted around the 1:1 line. The predicted SM for the assumed missing pixels kept high similarity with the SMAP official products, which thus have the potential to represent the SM status when the official SMAP SM was not available. For the OK method, the interpolation results in Fig. 9(b) based on the remaining 80% pixels correlated well with the 20% removed SMAP L3 SM products with a relatively high determinant coefficient of 0.85, but also with a high RMSE of 0.036 $\mathrm{cm^3/cm^3}$. The results indicated that the OK method captured the general trends of SMAP SM product, but with a larger deviation compared to RF.

C. Spatial Characteristics of Gap-Filling Soil Moisture

For different seasons over the Tibetan Plateau, Fig. 10 shows the gap-filling results using the RF and OK algorithms, respectively, in comparison with the original SMAP SM products.

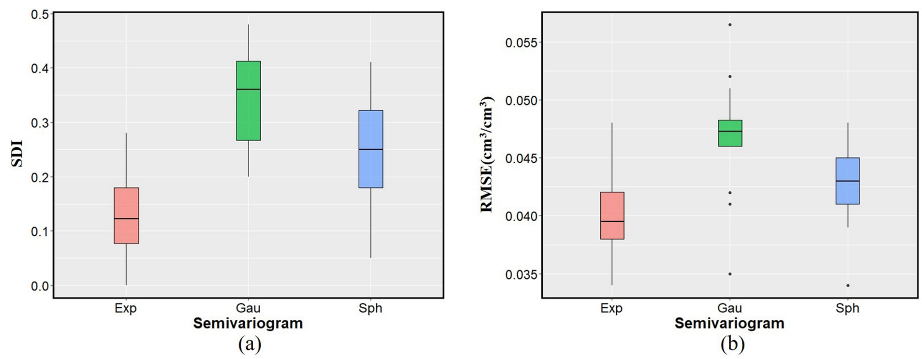

Fig. 8. Comparison of performances of three semivariograms in the OK algorithm in terms of (a) SDI (b) RMSE.

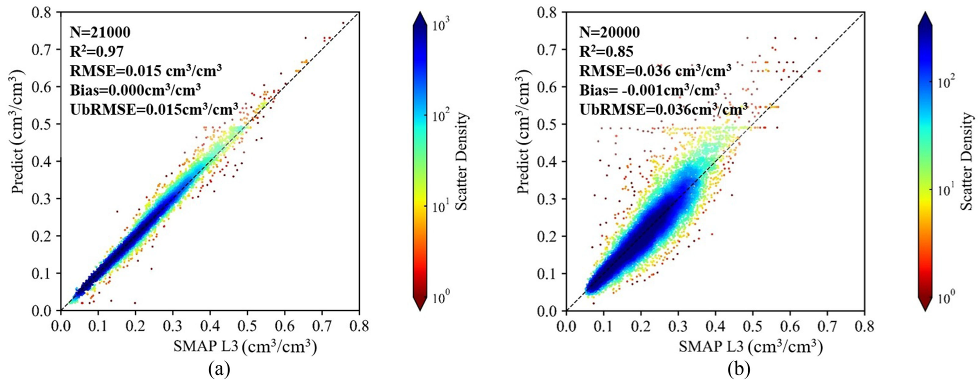

Fig. 9. Cross-validation for (a) Random Forest and (b) Ordinary kriging algorithms.

During different periods particularly when the temperature is low, the SMAP SM pixels were missing across different areas. The overall distribution of SM in the Tibetan Plateau is consistent, implying the eastern part is more humid than the western part that may be caused by altitude. The gap-filling results using the two algorithms present similar spatial patterns with the original SMAP SM. However, over the gap areas, the RF model retained spatial heterogeneity among different pixels with higher variance values, which is more close to the actual SM distribution. In contrast, the gap-filling results from the geostatistics method present smoother spatial distribution, thus this algorithm alleviates the differences between adjacent pixels. For some extremely high values, the interpolation results of the OK algorithm are smoothed and influenced by the neighboring pixels. In the southwest of the Tibetan Plateau from October 19th to October 21st, there was an extremely high SM pixel with considerable missing pixels around [Fig. 10(a)]. The gap-filling results from RF [Fig. 10(b)] maintain the heterogeneity with a high value surrounded by low values, while those from OK [Fig. 10(c)] show a gradient as the high value has gradients to all directions. Thus, compared with the geostatistics model, the RF model shows a better performance in reproducing SM heterogeneous patterns and capturing more spatial details.

D. Comparison With Other SM Data

Fig. 11 compares the gap-filling results from our proposed methods with the SMOS and GLDAS SM products for different seasons in 2018. The correlation coefficients between the gap-filling pixels and the available SMOS pixels were larger than 0.5. The SMAP and SMOS SM were retrieved using the similar τ–ω model applied to L-band radiometer signals [48], leading to similar sensed depth in the soil layers. Even for vegetation rapid growth seasons with high NDVI in Fig. 11(b) and (c), the gap-filling results for the SMAP SM data show a high agreement with the available SMOS SM data.

However, the gap-filling results using the two approaches were significantly lower than the GLDAS SM. This deviation may be attributed to the difference in the soil depths, as the GLDAS assimilation model simulates the SM at depth of 10 cm, while the SMAP retrieves the SM at depth of 5 cm. Although the GLDAS provided continuous and seamless SM data in time and space, their accuracy may be limited by uncertainties in a large number of input parameters such as atmospheric parameters, soil parameters, precipitation, and wind speed.

Furthermore, Fig. 12 shows the validation of the SMAP gap-filling results in terms of the *in-situ* SM over the Maqu network. However, the reconstructed SM is found to be systematically higher than the ground measured SM. This is due to the spatial heterogeneity of the SM which is challenging to be captured by limited number of ground stations. In addition, the scale difference between the *in-situ* network and the satellite footprint is another factor causing the bias. Nevertheless, the gap-filling results from RF ($R = 0.71$) and OK ($R = 0.55$) algorithms were significantly related to *in-situ* SM, indicating a certain agreement between the reconstructed and ground measured SM.

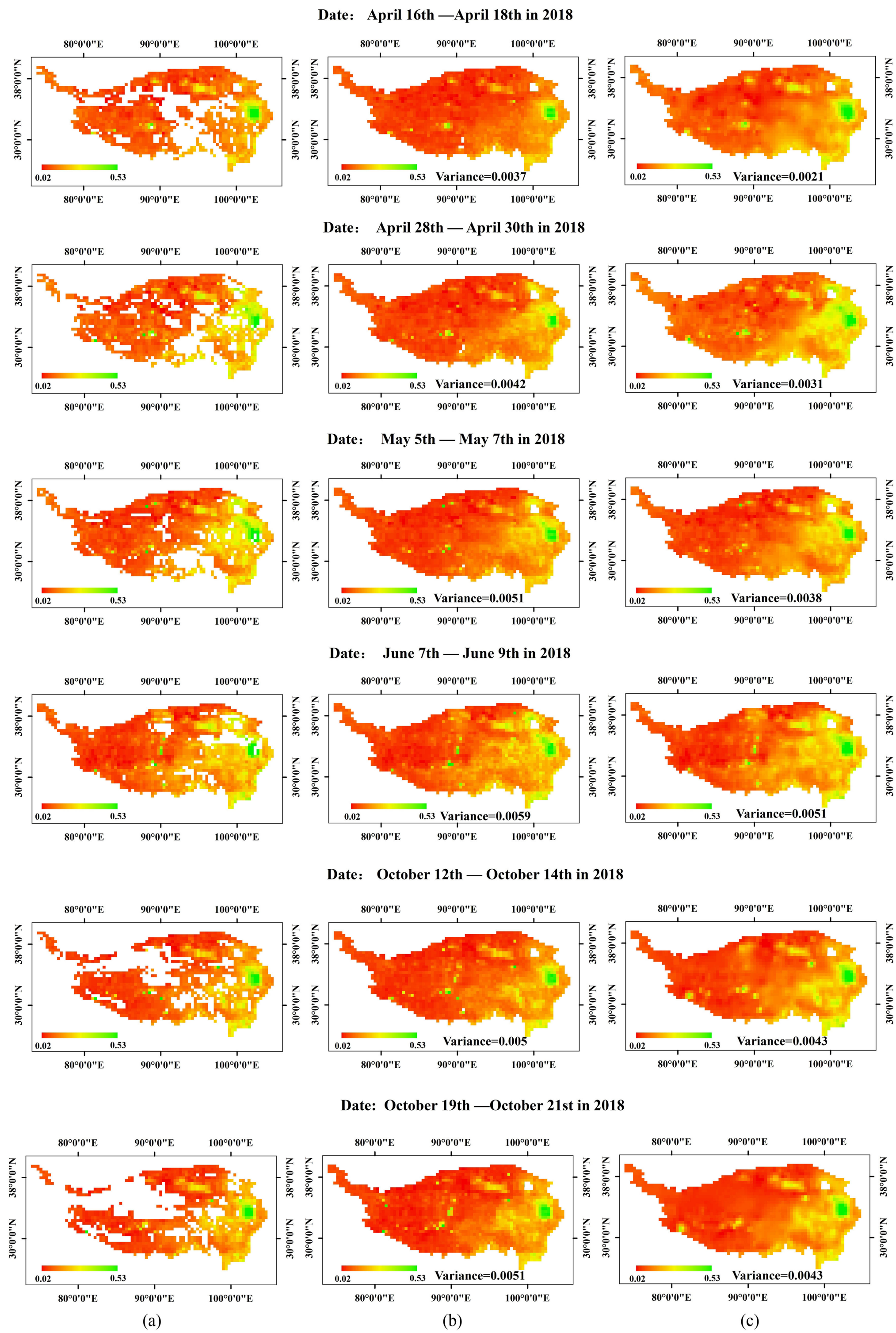

Fig. 10. (a) SMAP official soil moisture products in 2018 with issue of missing pxiels, in contrast to gap-filling results using (b) Random Forest and (c) Ordinary Kriging algorithms.

Compared with the OK method, the RF presented slightly better results in terms of statistical metrics. These comparisons reveal that the gap-filling pixels can be considered as reliable substitutes for the missing pixels in the satellite-derived SM products.

IV. Discussion

Due to satellite scanning gaps, dense vegetation, RFI contamination, and the limitation of physical conditions, the data missing of satellite-derived SM is an important issue. Past

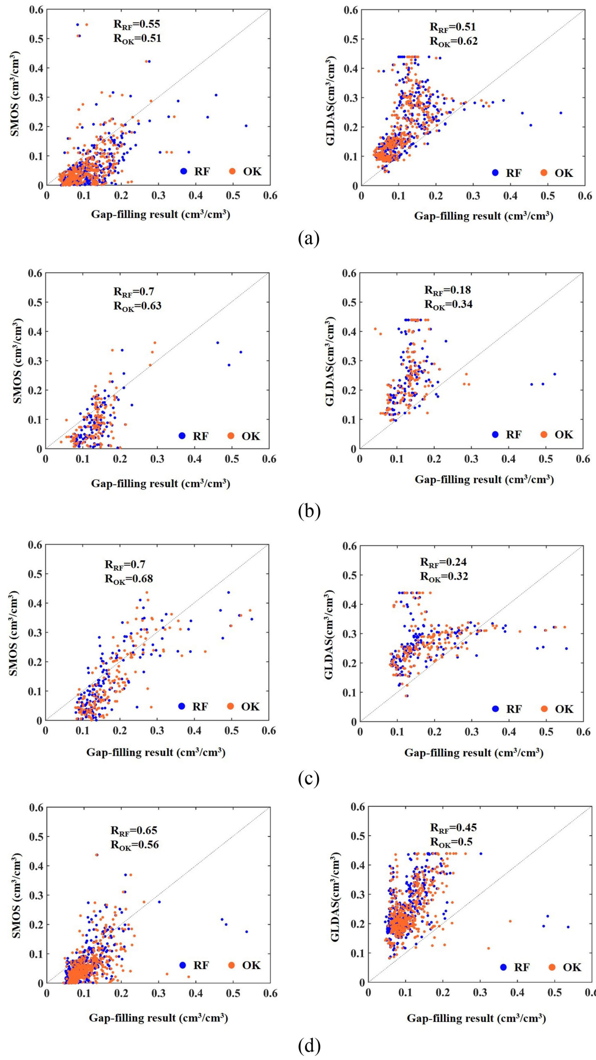

Fig. 11. Comparison of gap-filling SMAP SM results with SMOS and GLDAS soil moisture products in (a) April, (b) May, (c) June, and (d) October in 2018.

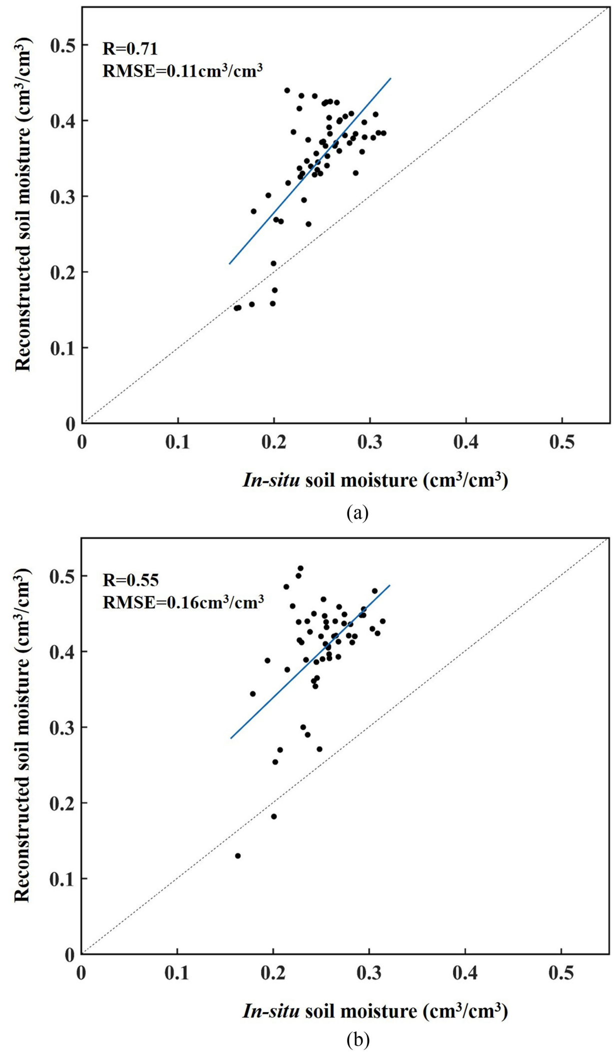

Fig. 12. Scatterplot of SMAP gap-filling results from (a) Random Forest and (b) Ordinary Kriging algorithms with *in-situ* soil moisture over Maqu network.

studies focused more on the ESA CCI SM [19], [20], [24], [49], [50] and paid less attention to SMAP product gaps. Over the Tibetan Plateau, the SMAP SM pixels were missing at varying degrees during different periods since 2015, which impedes their potential applications in environment, hydrology, climate, and data assimilation. Hence, in our article, we concentrated on this issue of SMAP SM pixels missing over the Tibetan Plateau and proposed two solutions to fill the missing pixel values and extend the spatial–temporal coverage of the SM data.

A. Inputs and Performances of Random Forest Algorithm

In case that the surface soil was unfrozen but SM value was missing, it was necessary to develop gap-filling approaches to reconstruct the missed pixels for obtaining complete SMAP SM data over Tibetan Plateau. In the current article, we evaluated the machine learning algorithm to establish a nonlinear relationship between several input variables and the SMAP SM. The ascending data were for training, and the missing pixels in descending orbit due to the failed retrieval from the physical radiative transfer model were reconstructed via the developed estimator. In comparison with [21] which evaluated the potential of the RF algorithm for SM retrieval over a semiarid agricultural region with a single land cover, the current article extends the algorithms to fill the SMAP SM missed pixels under complex land cover and high altitude conditions. Particularly, the performances of the gap-filling algorithm are compared to the traditional geostatistical approaches. Through the comparison with other SM products, the RF algorithm is found to be a potential solution to reconstruct the SMAP SM missed pixels with high robustness under diverse vegetation cover and high topography conditions. Thus, this algorithm improves the spatiotemporal coverage of SMAP SM data over TP. In addition, Qu *et al.* [51] investigated the RF algorithm to retrieve SM over the Tibetan Plateau from AMSR-E and AMSR2 brightness temperature, indicating a correlation $R = 0.94$ and RMSE $= 0.03$ m^3/m^3 with the SMAP SM. In contrast, our results were characterized by a slight higher correlation $R = 0.98$ and lower RMSE $= 0.015$ m^3/m^3.

For the RF algorithm, the appropriate selection of the input variables determines the performance. Through the comparison of the importance of input variables, we found that TB plays the most important role in reconstructing SMAP SM pixels. The spatial heterogeneity in TB leads to the retention of spatial details in the gap-filling SM. Meanwhile, our article selected the surface temperate as input, in agreement with the SMAP baseline algorithm. According to [52], the SMAP descending and ascending products based on τ–ω model used soil temperature to represent vegetation temperature. Thus, in our article, the corresponding ST in ascending and descending orbits was considered and applied in the training and reconstructing process. However, the heterogeneity in soil and vegetation characteristics in the evening was more complex than in the morning. It was anticipated that the accuracy of the descending SM is better than ascending SM, due to lower Faraday rotation and more uniform vegetation/soil temperature in the morning [53], [54].

In addition, NDVI significantly contributes to SM retrievals. Over the Tibetan Plateau, the vegetations including grassland, forest, and croplands were mainly distributed over the eastern part (Fig. 2), where our gap-filling results imply a higher SM status (Fig. 10). This confirmed the previous findings in [55] that SM was closely related to the MODIS NDVI value over the Tibetan Plateau, and the interaction between the SM and vegetation was a bidirectional process. This is in line with the use of vegetation index in optical remote sensing for SM retrieval and reconstruction [21], [50], [56], [57].

B. Advantages and Limitations of Ordinary Kriging Algorithm

The main advantage of the OK algorithm is that it did not need any auxiliary data and avoided the complex training process. The OK method satisfied Tobler's first law of geography which stated that spatial data values were more related to near values than distant ones [58]. In terms of SDI, SMAP pixels in Tibetan Plateau presented strong spatial variations. Thus, the valid pixels can be used to interpolate neighboring missing pixels. Compared with other interpolation techniques such as inverse distance weighting, the OK algorithm used the semivariogram function to calculate the weight of available SMAP SM pixels, which avoids the apparent noise situation in interpolation processes. However, the smoothing effect of the OK was considered as a drawback in geostatistics [59]. As an unbiased linear estimator, the OK method only considered spatial autocorrelation and ignored the issue of soil heterogeneity. Some extremely high or low values of SM may be regarded as noisy fields in interpolation processes and smoothed by adjacent values, which may cause underestimation or overestimation in regional areas. Moreover, the number of pixels applied in geostatistics technique has impacts on the filling results. The more sample points used in the study area, the greater performance of predicted values obtained. Thus, if there were many continuous missing pixels, or the valid pixels were below 50%, this method is not suitable for filling missing values.

Furthermore, in addition to the 36 km SMAP L3 SM product considered in this article, finer spatial resolution SMAP SM product (e.g., 9, 3, and 1 km) had also the issue of data gaps over the Tibetan Plateau. However, the coarse pixel alleviates the heterogeneity of neighboring pixels in comparison to finer pixels and represents a quasi-continuous matrix, which is more satisfying to the basic hypothesis of kriging interpolation such as spatial dependence and stationarity [24]. Meanwhile, given more heterogeneity in finer pixels, whether finer SMAP SM products can be reconstructed with favorable quality by the aforementioned machine learning algorithm is still an issue that needs to be studied in the following work.

C. Comparison of Gap-Filling Approaches for a Complete SMAP SM Data

By comparison of the two methods, our results indicated that the machine learning method with more input variables outperformed the geostatistics technique. However, in the extreme case when the satellite scanning gaps cover considerable continuous areas during a relatively long period, neither machine learning nor geostatistics method can be used. This happened when the SMAP satellite went into Safe Mode between 19 June and 23 July 2019 [26]. For this condition, the spatiotemporal characteristic of SM may be considered [60]. Thus, the potential applications of spatial–temporal Kriging [61] or spatiotemporal deep learning method [62] may be investigated to fill the SMAP data missing.

V. Conclusion

Spatial and temporal continuous SMAP SM data play an important role in various scientific researches and practical applications, especially over the Tibetan Plateau, which has significant effects on global climate change and water circle. However, it is found that significant gaps of SMAP SM appeared over Tibetan Plateau.

In this article, to solve the issue of data missing in SMAP 36 km SM L3 product over the Tibetan Plateau, we proposed the RF and OK algorithms to fill the SM gaps in the SMAP products. The machine learning RF method established a nonlinear relationship between input variables (SMAP TB_H, TB_V in ascending mode, ST, MODIS NDVI, auxiliary land cover, and DEM data) and the available SM pixels to reconstruct seamless SM maps over the Tibetan Plateau. The geostatistics OK technique takes the advantage of the spatial autocorrelation between existing pixels and missing values to interpolate complete SM data over the Tibetan Plateau. Three evaluation strategies are adopted to testify the accuracy of our gap-filling results: 1) cross-validation, 2) analysis on the spatial characteristics, and 3) intercomparison with independent SM samples. The validation results show that the gap-filling SM has a high similarity to the SMAP official products and with strong robustness.

In summary, both the machine learning method and geostatistics technique have the potentials to improve the temporal and spatial coverages of the SMAP SM products, and may break out the limitations of retrievals from the traditional radiative transfer models. However, compared with the geostatistics technique, the RF method performed better and captured more spatial details, due to the added information from more input variables. Moreover, the ST was found to be the major reason causing the data missing over the Tibetan Plateau. In perspective, two aspects of gap-filling SMAP are worth considering: one is to reconstruct global seamless SMAP products, and the other is to apply the gap-filling methods to SMAP products at higher spatial resolution (e.g., 9, 3, and 1 km).

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